CISCO PRACTICAL – FINAL MASTER CHEAT SHEET

1. IP CONFIG SPECIFICATIONS

Standard Format

```
Network: 192.168.1.0
Mask: 255.255.255.0
PC1: 192.168.1.2
PC2: 192.168.1.3
Gateway: 192.168.1.1
```

2. CABLING

- PC1 console → console
- Switch console → console
- Switch Fa0/1 → router
- Router console → console

3. COMMON COMMANDS

```
enable
configure terminal

! Set hostname
hostname R1

! Enter interface
interface Fa0/0
! Bring up and set internal IP address
ip address 192.168.1.1 255.255.255.0 ! ip address ip_address (subnet mask)
! Bring up the interface
no shutdown

! Save
copy running-config startup-config
show ip route

show ip interface brief

! Save the current configuration
copy running-config startup-config
write
write memory
```

4. ASSIGNMENT 1 (BASIC + HUB + SWITCH + ROUTER)

Steps

- Create topology.
- Assign IP addresses.
- Configure PC1 and PC2 in Configuration.
- Ping.

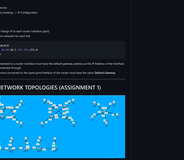
For Routers

- Enable and Assign IP to each router interface (port).
- Use different networks for each link.

```
interface Fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

- Devices connected to a router interface must have the default gateway address as the IP Address of the interface they are connected through.
- For end devices connected to the same port/interface of the router must have the same **Default Gateway**.

5. NETWORK TOPOLOGIES (ASSIGNMENT 1)



Types

- Star
- Bus
- Ring
- Mesh

Steps

- Place PCs and switches.
- Connect with cable topology.
- Assign IPs.
- Ping between PCs.

6. SWITCH CONFIG (ASSIGNMENT 2)

```
enable
conf t

hostname Switch1

banner MOTD #Unauthorized access#

line console 0
password cisco
login
exec-timeout 60 0
logging synchronous

! Save
copy running-config startup-config
write
write memory

exit

Then Reboot the Switch
```

7. VLAN CONFIG (ASSIGNMENT 3)

IP CONFIG TABLE

Device	VLAN	Port Assignment	IP Address	Subnet Mask	Gateway
PC1	10	FastEthernet0/1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	10	FastEthernet0/2	192.168.1.3	255.255.255.0	192.168.1.1
PC3	20	FastEthernet0/3	192.168.2.2	255.255.255.0	192.168.2.1
PC4	20	FastEthernet0/4	192.168.2.3	255.255.255.0	192.168.2.1

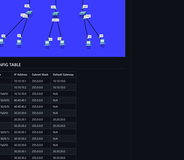
Switch Commands

```
enable
conf t

! Create VLANs
vlan 10
name VLAN_10
exit

vlan 20
name VLAN_20
exit

interface range Fa0/1-2
switchport mode access
switchport access vlan 10
exit

interface range Fa0/3-4
switchport mode access
switchport access vlan 20
exit

interface Fa0/5
switchport mode trunk
exit

! Save
copy running-config startup-config
write
write memory
```

Router Commands

```
enable
conf t

int Fa0/0
no shutdown
exit

int Fa0/0.10
encapsulation dot1Q 10
ip address 192.168.1.1 255.255.255.0
! ip address ip_address subnet_mask (mask in size 24: (subnet mask))

int Fa0/0.20
encapsulation dot1Q 20
ip address 192.168.2.1 255.255.255.0
! ip address ip_address subnet_mask (mask in size 24: (subnet mask))

! Save
copy running-config startup-config
write
write memory
```

8. STATIC ROUTING (ASSIGNMENT 4)



IP CONFIG TABLE

Device	IP Address	Subnet Mask	Default Gateway
R1(S)	10.10.10.1	255.0.0.0	10.10.10.2
R1(C)	10.10.10.2	255.0.0.0	10.10.10.1
Router1 (Fa1/0/0)	10.10.10.10	255.0.0.0	N/A
Router1 (Fa1/0/1)	20.20.20.1	255.0.0.0	N/A
Router2 (Fa1/0/0)	20.20.20.2	255.0.0.0	N/A
Router2 (Fa1/0/1)	30.30.30.1	255.0.0.0	N/A
Router3 (Fa1/0/0)	30.30.30.2	255.0.0.0	N/A
Router3 (Fa1/0/1)	40.40.40.1	255.0.0.0	N/A
PC1	10.10.10.3	255.0.0.0	10.10.10.1
PC2	20.20.20.3	255.0.0.0	20.20.20.1
PC3	30.30.30.3	255.0.0.0	30.30.30.1
PC4	40.40.40.3	255.0.0.0	40.40.40.1

CLI

```
enable
conf t
! Enable static routing
ip route 20.20.20.0 255.255.255.0 10.10.10.1
ip route 30.30.30.0 255.255.255.0 20.20.20.2
ip route 40.40.40.0 255.255.255.0 30.30.30.2

! Save
copy running-config startup-config
write
write memory
```

- Create topology.
- Assign IP address to PCs.
- Enable static routing on routers.
- Ping.

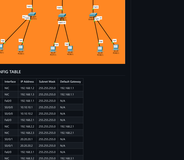
For Routers

- Enable and Assign IP to each router interface (port).
- Use different networks for each link.

```
interface Fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

- Devices connected to a router interface must have the default gateway address as the IP Address of the interface they are connected through.
- For end devices connected to the same port/interface of the router must have the same **Default Gateway**.

9. RIP ROUTING (ASSIGNMENT 5)



IP CONFIG TABLE

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1(C)	Fa0	192.168.0.1	255.255.255.0	192.168.0.2
R1(S)	Fa0	192.168.0.2	255.255.255.0	192.168.0.1
Router 1 (PT)	R0/0/0	10.10.10.1	255.255.255.0	N/A
Router 1 (PT)	R0/0/1	50.50.50.1	255.255.255.0	N/A
Router 2 (PT)	R0/0/0	10.10.10.2	255.255.255.0	N/A
Router 2 (PT)	R0/0/1	20.20.20.1	255.255.255.0	N/A
Router 3 (PT)	R0/0/0	20.20.20.2	255.255.255.0	N/A
Router 3 (PT)	R0/0/1	192.168.0.2	255.255.255.0	192.168.0.1
PC1	NIC	192.168.0.3	255.255.255.0	192.168.0.1
Server 1 (PT)	R0/0/1	20.20.20.1	255.255.255.0	N/A
Server 2 (PT)	R0/0/1	20.20.20.2	255.255.255.0	N/A
Server 3 (PT)	R0/0/1	192.168.0.2	255.255.255.0	N/A
Server 4 (PT)	R0/0/1	192.168.0.1	255.255.255.0	N/A
PC2	NIC	192.168.0.4	255.255.255.0	192.168.0.1
PC3	NIC	192.168.0.5	255.255.255.0	192.168.0.1

CLI

```
router rip
! to add network of the network on the router
network 192.168.0.0

! add the address of the remote serial port the address the router itself is discovered
network 50.50.50.0
```

UI Steps

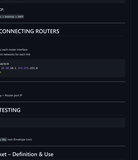
- Create topology.
- Assign IP.
- Add networks.

10. DHCP USING SERVER

Steps

- Server → Desktop → IP Config
  - IP: 192.168.1.1
- Services → DHCP
  - Default Gateway: 192.168.1.1
  - Start IP address: 192.168.1.0
- PCs → Desktop → DHCP

11. DHCP USING ROUTER



CLI Commands

```
enable
configure terminal

interface Fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface Fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

! Enable DHCP Server

! Step excluded address 192.168.1.1
ip dhcp excluded-address 192.168.1.1

! Step excluded address 192.168.1.2
ip dhcp excluded-address 192.168.1.2

! Step pool Test1
ip dhcp pool Test1
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1

! Step pool Test2
ip dhcp pool Test2
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1

! Save
copy running-config startup-config
write
write memory
```

Use DHCP

Pass Phrase: **Mr.Robot#Rules-1-Mr#**

12. CONNECTING ROUTERS

Steps:

- Assign IP to each router interface
- Use different networks for each link

```
interface Fa0/0
ip address 10.10.10.1 255.255.255.0
no shutdown
```

Rule:

- PC Gateway = Router port IP

13. TESTING

```
ping <IP>

ok
```

See Cisco Tiki Test Broadcast Icon

14. Socket – Definition & Use

A socket is a software abstraction that represents a network endpoint, combining an IP address and port number, used for communication between two devices over a network using the TCP or UDP protocol.

15. Basic TCP Socket (Connect to Server)

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect("www.google.co.in", 80)
print("Connected to www.google.co.in on port 80")
s.close()
```

- `socket` – Core TCP socket
- `connect()` – establishes connection to server
- `close()` – Frees connection

16. TCP Server

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_socket.bind(("0.0.0.0", 12345))
server_socket.listen(1)

client_socket, addr = server_socket.accept()
data = client_socket.recv(1024).decode()
client_socket.send("Hello Client".encode())

client_socket.close()
server_socket.close()
```

- `bind()` – attaches IP & port
- `listen()` – waits for clients
- `accept()` – accepts connection
- `recv()` – receives data
- `send()` – sends data

17. TCP Client

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect("127.0.0.1", 12345)

client_socket.send("Hello Server".encode())
response = client_socket.recv(1024).decode()

client_socket.close()
```

- `connect()` – connects to server
- `send()` – sends message
- `recv()` – receives response

18. UDP Server

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server_socket.bind(("0.0.0.0", 12345))

data, addr = server_socket.recvfrom(1024)
server_socket.sendto("Hello Client".encode(), addr)

server_socket.close()
```

- `SOCK_DGRAM` – UDP socket
- `recvfrom()` – receives datagram + address
- `sendto()` – send data to client

19. UDP Client

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

client_socket.sendto("Hello UDP Server".encode(), ("127.0.0.1", 12345))
response, addr = client_socket.recvfrom(1024)

client_socket.close()
```

- `sendto()` – sends data without connection
- `recvfrom()` – receives response

FINAL QUICK MEMORY

- TCP = connection-based
- UDP = connectless
- Server → bind → listen → accept
- Client → connect → send → receive